Sleep Breathing Irregularity Detection — Project Report

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1. Introduction

Sleep breathing irregularities such as Hypopnea and Obstructive Apnea are serious medical conditions that reduce oxygen supply during sleep. This project builds an automated detection pipeline using physiological signals collected from 5 participants over 8-hour overnight sleep sessions.

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2. Dataset

Data was collected from 5 participants (AP01–AP05). Each participant folder contains:

- **Nasal Airflow** — sampled at 32 Hz
- **Thoracic Movement** — sampled at 32 Hz
- **SpO2 (Oxygen Saturation)** — sampled at 4 Hz
- **Flow Events file** — annotated breathing irregularities with start time, end time, and label
- **Sleep Profile file** — sleep stage annotations every 30 seconds

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3. Visualization

For each participant, all 3 signals were plotted across the full 8-hour recording on a shared time axis. Annotated events were overlaid as colored bands — orange for Hypopnea and red for Obstructive Apnea. Key observation: events were concentrated between hours 3–6, coinciding with deeper sleep stages. SpO2 showed visible dips during apnea events, confirming biological validity of the annotations.

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4. Signal Preprocessing

Filtering: A 4th order Butterworth bandpass filter was applied to all signals retaining only the 0.17–0.4 Hz frequency range corresponding to normal human breathing (10–24 breaths per minute). This removed high frequency noise and movement artifacts.

Windowing: Signals were segmented into 30-second windows with 50% overlap (15-second step). This produced:

- Flow and Thorac windows: 960 samples each
- SpO2 windows: 120 samples each (upsampled to 960 during modeling)

Labeling: Each window was labeled based on overlap with annotated events. If a window overlapped more than 50% with a labeled event, it received that event's label. Otherwise it was labeled Normal.

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5. Dataset Statistics

Label	Windows
Normal	8041
Hypopnea	593
Obstructive Apnea	163
Total	**8797**

The dataset shows significant class imbalance which is expected and typical in clinical sleep studies.

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6. Model Architecture

A 1D CNN was designed to classify 30-second windows into 3 classes. The architecture consists of:

- 3 Conv1D blocks with increasing filters (32, 64, 128) and kernel sizes (5, 5, 3)
- BatchNormalization after each conv block for training stability
- GlobalAveragePooling to reduce spatial dimensions
- Dense layer (64 units) with 0.3 Dropout for regularization
- Softmax output layer for 3-class classification

Total parameters: 43,971 — lightweight and fast to train.

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7. Evaluation Strategy

Leave-One-Participant-Out Cross Validation was used. In each of the 5 folds, the model was trained on 4 participants and tested on the remaining one. Class weights were applied during training to handle class imbalance, penalizing the model more for missing abnormal events.

8. Results

Participant	Accuracy	Precision	Recall
AP01	0.4292	0.3668	0.6206
AP02	0.7575	0.3850	0.6212
AP03	0.1374	0.3308	0.3794
AP04	0.7969	0.3544	0.3672
AP05	0.4304	0.4002	0.4924
Average	**0.5103**	**0.3674**	**0.4962**

9. Discussion

AP02 and AP04 achieved good accuracy (75–80%) due to balanced event distribution. AP03 performed poorly (14%) because it had only 17 abnormal windows out of 1696 total — making generalization nearly impossible. The low precision across all folds reflects the inherent difficulty of cross-patient generalization with only 5 participants.

10. Conclusion

This project successfully built an end-to-end pipeline for sleep breathing irregularity detection — from raw signal parsing to CNN classification. Despite the small dataset size and class imbalance, the model detected abnormal breathing events with reasonable recall. Performance would improve significantly with more participants and advanced architectures such as Conv-LSTM.